

NETWORK TRAFFIC ANALYSIS

Malware Delivery & DNS Tunneling Investigation

Incident Reference: BulbaTech Innovations — Abnormal Endpoint Traffic

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Tool Used: Wireshark

Date: June 30, 2026

1. Executive Summary

The SOC team at BulbaTech Innovations identified abnormal traffic patterns and a high volume of repeated DNS queries originating from an external-facing endpoint (172.16.1.16). A full packet capture (wireshark_challenge.pcap) was provided for forensic analysis.

Using Wireshark, the capture was analyzed end-to-end to identify the initial point of compromise, the delivery mechanism of the malicious payload, and ongoing command-and-control (C2) activity. The investigation confirmed that the endpoint was compromised via a disguised file download and is exhibiting signs of DNS tunneling, consistent with Qakbot trojan behavior.

Affected Host	172.16.1.16
Malware Identified	Trojan/Win.Qakbot
Delivery Method	HTTP download disguised as a GIF image (actual file: Windows PE executable)
Post-Infection Activity	Sustained, repeated DNS queries to a single base domain — consistent with DNS Tunneling (MITRE ATT&CK T1071.004)
Total Packets Analyzed	39,106

2. Scope & Tools

2.1 Objective

Analyze the provided packet capture to determine how the endpoint was compromised, what payload was delivered, and what ongoing malicious behavior is present on the network.

2.2 Evidence Provided

- File: wireshark_challenge.pcap
- Total packets: 39,106
- Suspected compromised host: 172.16.1.16

2.3 Tools Used

- Wireshark — packet capture analysis, display filters, Follow HTTP Stream
- sha256sum (Linux CLI) — file hashing
- VirusTotal — threat intelligence lookup

3. Analysis & Findings

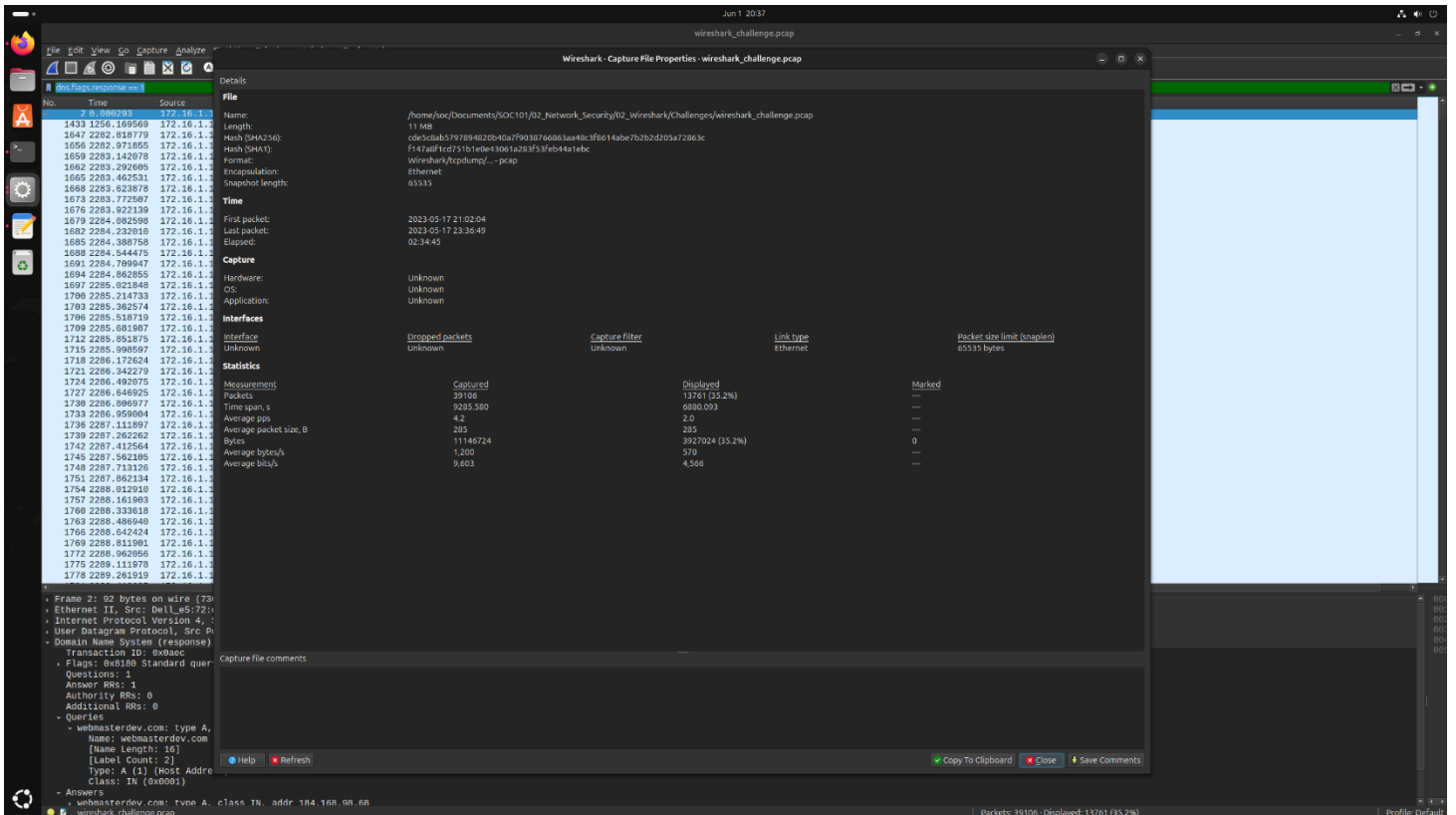
Each question below documents the investigative step taken, the Wireshark filter or technique used, and the resulting finding. Screenshot placeholders are provided — insert your corresponding evidence screenshot directly below each finding.

3.1 Capture Overview

Q1. How many total packets are in the wireshark_challenge.pcap packet capture?

Method / Filter: Statistics → Capture File Properties

Finding: 39,106 total packets.



3.2 Initial DNS Resolution

Q2. What was the first domain name queried and resolved in the capture?

Method / Filter: dns.flags.response == 1

Finding: webmasterdev.com

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wireshark_challenge.pcap

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dns.flags.response == 1

No.	Time	Source	Destination	Protocol	Length	Info
2	0.000000	172.16.1.191	172.16.1.101	DNS	132	Standard query response 0x58ac A:webmasterdev.com A:184.168.98.64
1433	12.506160	172.16.1.191	172.16.1.101	DNS	132	Standard query response 0x3070 A:twitter.com A:104.244.42.1 A:104.244.42.65 A:104.244.42.103 A:104.244.42.129
1647	22.821877	172.16.1.191	172.16.1.101	DNS	351	Standard query response 0x6394 TXT:aaa.h.dns.steassteel.net TXT
1696	22.821918	172.16.1.191	172.16.1.101	DNS	351	Standard query response 0xb220 TXT:baa.h.dns.steassteel.net TXT
1698	22.831428	172.16.1.191	172.16.1.101	DNS	351	Standard query response 0x2c6f TXT:caa.h.dns.steassteel.net TXT
1662	22.832605	172.16.1.191	172.16.1.101	DNS	351	Standard query response 0xf622 TXT:daa.h.dns.steassteel.net TXT
1685	22.834823	172.16.1.191	172.16.1.101	DNS	351	Standard query response 0x809e TXT:aaa.h.dns.steassteel.net TXT
1686	22.834938	172.16.1.191	172.16.1.101	DNS	351	Standard query response 0xc57e TXT:faa.h.dns.steassteel.net TXT
1673	22.837750	172.16.1.191	172.16.1.101	DNS	351	Standard query response 0x140a TXT:ga.h.dns.steassteel.net TXT
1676	22.838213	172.16.1.191	172.16.1.101	DNS	351	Standard query response 0xd770 TXT:haa.h.dns.steassteel.net TXT
1678	22.841823	172.16.1.191	172.16.1.101	DNS	351	Standard query response 0xcacaa TXT:iaa.h.dns.steassteel.net TXT
1682	22.842320	172.16.1.191	172.16.1.101	DNS	351	Standard query response 0x49be TXT:jaa.h.dns.steassteel.net TXT
1685	22.843878	172.16.1.191	172.16.1.101	DNS	351	Standard query response 0xf8be TXT:kaa.h.dns.steassteel.net TXT
1686	22.844475	172.16.1.191	172.16.1.101	DNS	351	Standard query response 0x8536 TXT:laa.h.dns.steassteel.net TXT
1691	22.847694	172.16.1.191	172.16.1.101	DNS	351	Standard query response 0xc1c7 TXT:maa.h.dns.steassteel.net TXT
1694	22.848625	172.16.1.191	172.16.1.101	DNS	351	Standard query response 0x7a75 TXT:naa.h.dns.steassteel.net TXT
1697	22.851014	172.16.1.191	172.16.1.101	DNS	351	Standard query response 0xe7c3 TXT:oa.h.dns.steassteel.net TXT
1706	22.852147	172.16.1.191	172.16.1.101	DNS	351	Standard query response 0xffb1 TXT:pa.h.dns.steassteel.net TXT
1703	22.852624	172.16.1.191	172.16.1.101	DNS	351	Standard query response 0x4d40 TXT:qaa.h.dns.steassteel.net TXT
1706	22.853173	172.16.1.191	172.16.1.101	DNS	351	Standard query response 0x98a5 TXT:raa.h.dns.steassteel.net TXT
1708	22.856187	172.16.1.191	172.16.1.101	DNS	351	Standard query response 0x5af0 TXT:sa.h.dns.steassteel.net TXT
1712	22.858187	172.16.1.191	172.16.1.101	DNS	351	Standard query response 0x0657 TXT:taa.h.dns.steassteel.net TXT
1715	22.859087	172.16.1.191	172.16.1.101	DNS	351	Standard query response 0x3d43 TXT:uaa.h.dns.steassteel.net TXT
1718	22.861726	172.16.1.191	172.16.1.101	DNS	351	Standard query response 0x4437 TXT:vaa.h.dns.steassteel.net TXT
1721	22.863427	172.16.1.191	172.16.1.101	DNS	351	Standard query response 0xb220 TXT:wa.h.dns.steassteel.net TXT
1724	22.864925	172.16.1.191	172.16.1.101	DNS	351	Standard query response 0x5011 TXT:xaa.h.dns.steassteel.net TXT
1727	22.866492	172.16.1.191	172.16.1.101	DNS	351	Standard query response 0xe07f TXT:yaa.h.dns.steassteel.net TXT
1738	22.868697	172.16.1.191	172.16.1.101	DNS	351	Standard query response 0xe0b0 TXT:zaa.h.dns.steassteel.net TXT
1733	22.869086	172.16.1.191	172.16.1.101	DNS	351	Standard query response 0xc4c1 TXT:ab.h.dns.steassteel.net TXT
1736	22.871187	172.16.1.191	172.16.1.101	DNS	351	Standard query response 0xc116 TXT:bba.h.dns.steassteel.net TXT
1739	22.872622	172.16.1.191	172.16.1.101	DNS	351	Standard query response 0x1666 TXT:caa.h.dns.steassteel.net TXT
1742	22.874124	172.16.1.191	172.16.1.101	DNS	351	Standard query response 0xc4cd TXT:dna.h.dns.steassteel.net TXT
1745	22.876105	172.16.1.191	172.16.1.101	DNS	351	Standard query response 0xd0fa TXT:eba.h.dns.steassteel.net TXT
1748	22.877312	172.16.1.191	172.16.1.101	DNS	351	Standard query response 0xe586 TXT:faa.h.dns.steassteel.net TXT
1751	22.878624	172.16.1.191	172.16.1.101	DNS	351	Standard query response 0x068f TXT:ga.h.dns.steassteel.net TXT
1754	22.880120	172.16.1.191	172.16.1.101	DNS	351	Standard query response 0xa90d TXT:hba.h.dns.steassteel.net TXT
1757	22.881610	172.16.1.191	172.16.1.101	DNS	351	Standard query response 0x35d0 TXT:iba.h.dns.steassteel.net TXT
1766	22.883391	172.16.1.191	172.16.1.101	DNS	351	Standard query response 0xb417 TXT:jba.h.dns.steassteel.net TXT
1763	22.884604	172.16.1.191	172.16.1.101	DNS	351	Standard query response 0x7317 TXT:kba.h.dns.steassteel.net TXT
1766	22.886424	172.16.1.191	172.16.1.101	DNS	351	Standard query response 0xd9a7 TXT:lba.h.dns.steassteel.net TXT
1768	22.888190	172.16.1.191	172.16.1.101	DNS	351	Standard query response 0xc97c TXT:mha.h.dns.steassteel.net TXT
1772	22.889626	172.16.1.191	172.16.1.101	DNS	351	Standard query response 0x899c TXT:nba.h.dns.steassteel.net TXT
1775	22.891197	172.16.1.191	172.16.1.101	DNS	351	Standard query response 0xa5b5 TXT:oba.h.dns.steassteel.net TXT
1778	22.892618	172.16.1.191	172.16.1.101	DNS	351	Standard query response 0x1510 TXT:pba.h.dns.steassteel.net TXT

Frame 2: 92 bytes on wire (736 bits), 92 bytes captured (736 bits) on interface 0
 Ethernet II, Src: Dell-e572ef (54:00:14:00:00:00), Dst: Hewlett-Packard-e572ef (54:00:14:00:00:00)
 Internet Protocol Version 4, Src: 172.16.1.191, Dst: 172.16.1.101
 User Datagram Protocol, Src Port: 53, Dst Port: 51176
 Domain Name System (Response)
 Transaction ID: 0x58ac
 Flags: 0x0100 Standard query response, No error
 Questions: 1
 Answer RRs: 1
 Authority RRs: 0
 Additional RRs: 0
 Queries:
 webmasterdev.com: type A, class IN
 Name: webmasterdev.com
 (Name Length: 16)
 (Label Count: 2)
 Type: A (1) (Host Address)
 Class: IN (0x0001)
 Answers:
 webmasterdev.com: type A, class IN, addr 184.168.98.64
 wireshark_challenge.pcap

Packets: 39106 - Displayed: 13761 (35.2%)

Q3. What is the associated IP address of the domain name?

Method / Filter: dns.flags.response == 1 (same filtered packet, DNS Answer section)

Finding: 184.168.98.64

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wireshark_challenge.pcap

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ip.addr == 184.168.98.68 and tls

No.	Time	Source	Destination	Protocol	Length	Info
6	0.492764	184.168.98.68	172.16.1.191	TLSv1.2	1514	Server Hello
13	0.624758	184.168.98.68	172.16.1.191	TLSv1.2	1420	Certificate, Certificate Status
14	0.624938	184.168.98.68	172.16.1.191	TLSv1.2	77	Server Key Exchange, Server Hello Done
16	0.624446	172.16.1.191	184.168.98.68	TLSv1.2	147	Client Key Exchange, Change Cipher Spec, Encrypted Handshake Message
18	0.625808	172.16.1.191	184.168.98.68	TLSv1.2	159	Application Data
20	0.628738	172.16.1.191	184.168.98.68	TLSv1.2	541	Application Data
22	0.666526	184.168.98.68	172.16.1.191	TLSv1.2	344	New Session Ticket, Change Cipher Spec, Encrypted Handshake Message
23	0.666957	184.168.98.68	172.16.1.191	TLSv1.2	111	Application Data
25	0.667165	172.16.1.191	184.168.98.68	TLSv1.2	92	Application Data
27	1.107699	184.168.98.68	172.16.1.191	TLSv1.2	92	Application Data
42	0.628083	184.168.98.68	172.16.1.191	TLSv1.2	1514	Application Data
61	3.028627	184.168.98.68	172.16.1.191	TLSv1.2	1430	Application Data
79	3.270225	184.168.98.68	172.16.1.191	TLSv1.2	1514	Application Data
93	3.379608	184.168.98.68	172.16.1.191	TLSv1.2	1094	Application Data
106	3.511337	184.168.98.68	172.16.1.191	TLSv1.2	255	Application Data
108	8.377100	184.168.98.68	172.16.1.191	TLSv1.2	138	Application Data, Encrypted Alert

Frame 6: 571 bytes on wire (4568 bits), 571 bytes captured (4568 bits) on interface 0
 Ethernet II, Src: Hewlett-Packard-e572ef (54:00:14:00:00:00), Dst: Cisco_7a:1d:39 (08:00:c7:7a:1d:39)
 Internet Protocol Version 4, Src: 172.16.1.191, Dst: 184.168.98.68
 0100 ... = Version: 4
 ... 0101 = Header Length: 20 bytes (5)
 - Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
 0000 0000 ... = Differentiated Services Codepoints: Default (0)
 ... 00 = Explicit Congestion Notification: Not ECN-capable Transport (0)
 Total Length: 567
 Identifications: 0x0000 (63454)
 0100 ... = Flags: 0x02, Don't Fragment
 0 ... = Reserved bit: Not set
 1 ... = Don't Fragment: Set
 0 ... = More fragments: Not set
 ... 0000 0000 0000 = Fragment Offset: 0
 Time to Live: 120
 Protocol: TCP (6)
 Header Checksum: 0x5f40 [validation disabled]
 (Header checksum status: Unverified)
 Source Address: 172.16.1.191
 Transport Layer Security Protocol

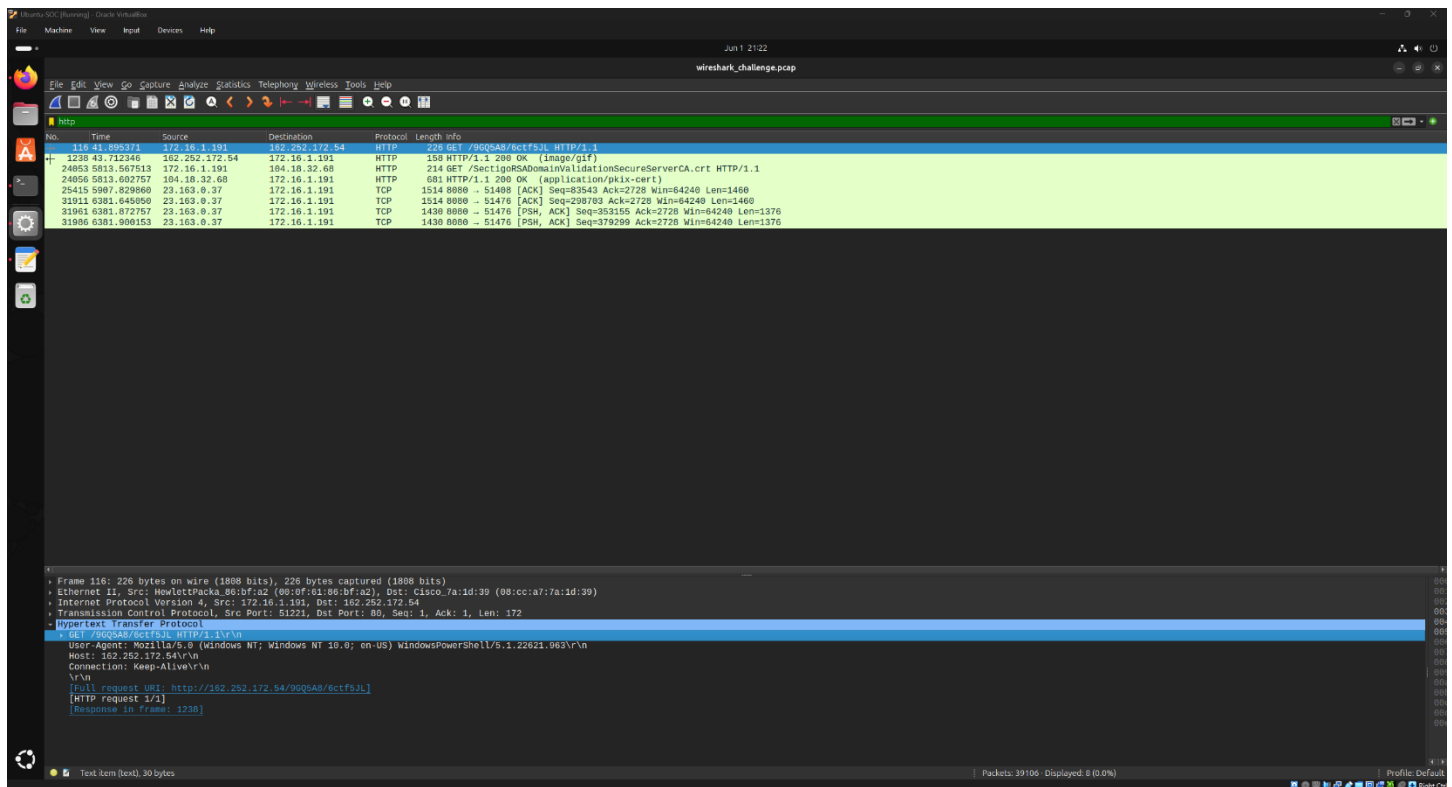
Packets: 39106 - Displayed: 17 (0.0%)

3.3 HTTP Payload Delivery

Q4. How many HTTP packets are contained in the capture file?

Method / Filter: http

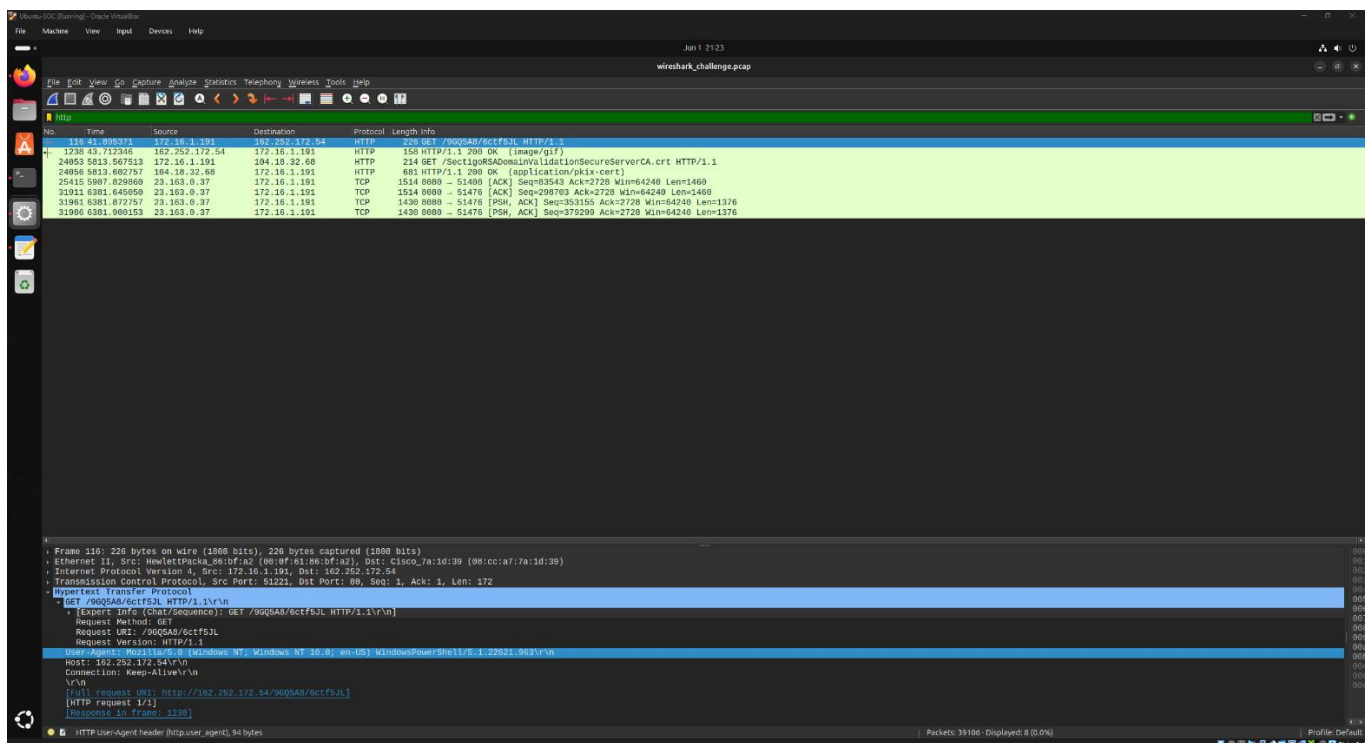
Finding: 8 HTTP packets total.



Q5. What is the relative path the victim accessed on the web server to request a file for download?

Method / Filter: Examined the HTTP GET request header

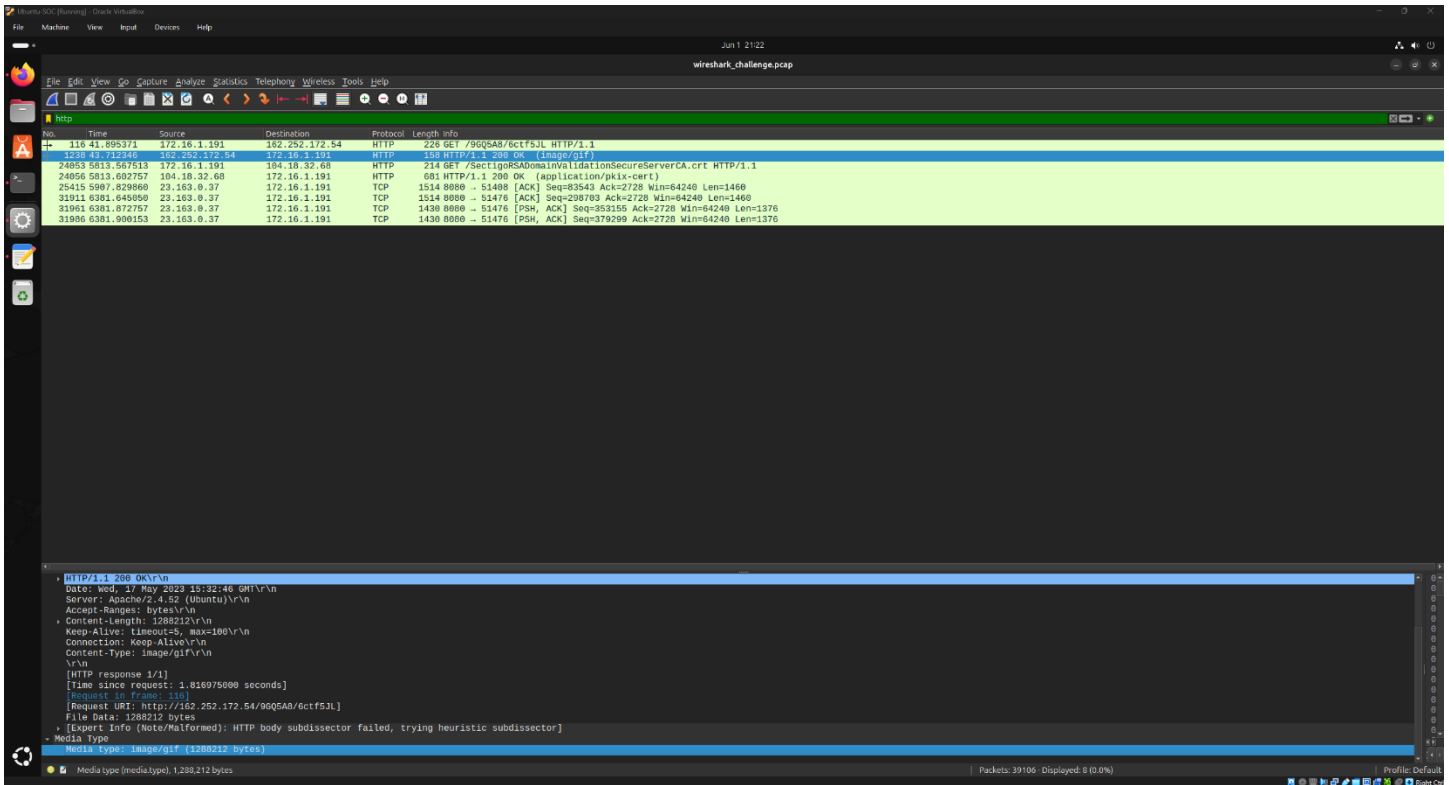
Finding: /9GQ5A8/6ctf5JL



Q6. Based on the response header, what file type or format does the web server claim the downloaded file to be?

Method / Filter: Follow → HTTP Stream on the request/response pair

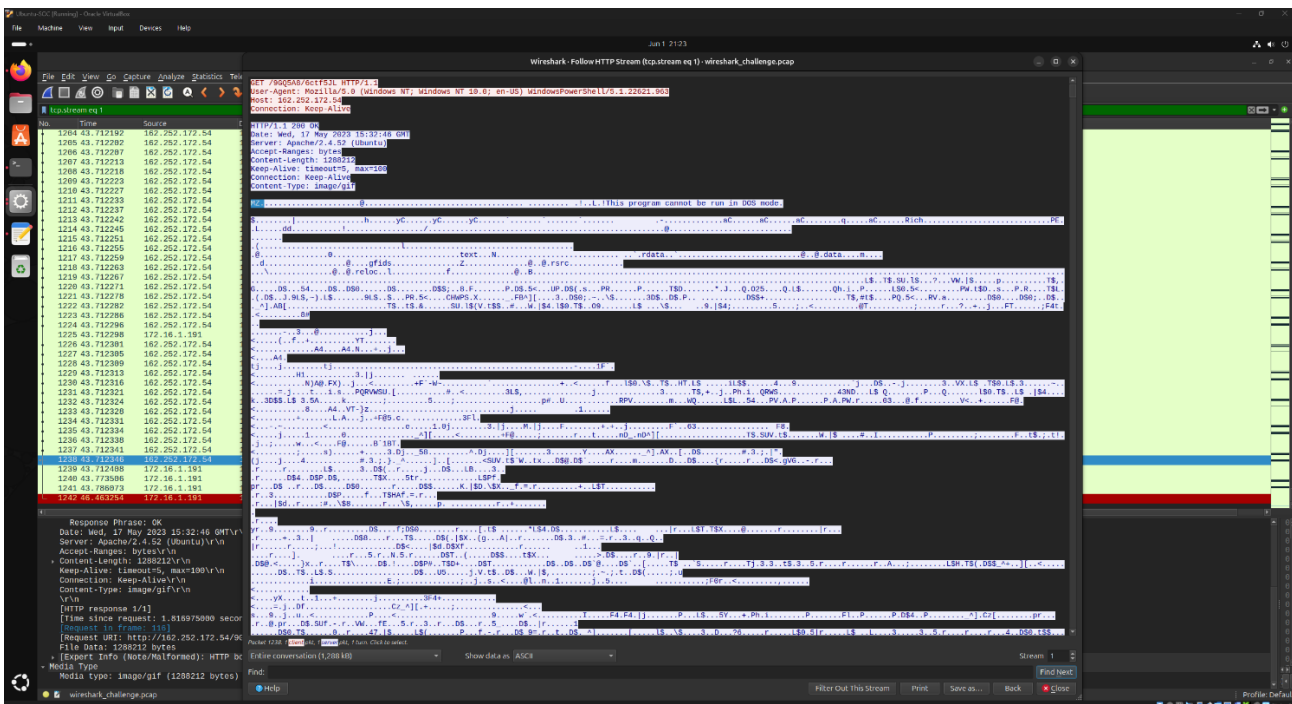
Finding: The server's Content-Type header claims the file is image/gif.



Q7. However, what is the actual file signature or magic bytes contained in the file?

Method / Filter: Inspected the raw file bytes exported from the HTTP stream

Finding: The file signature is MZ — the magic bytes for a Windows Portable Executable (PE), confirming the file is a disguised executable, not a GIF.

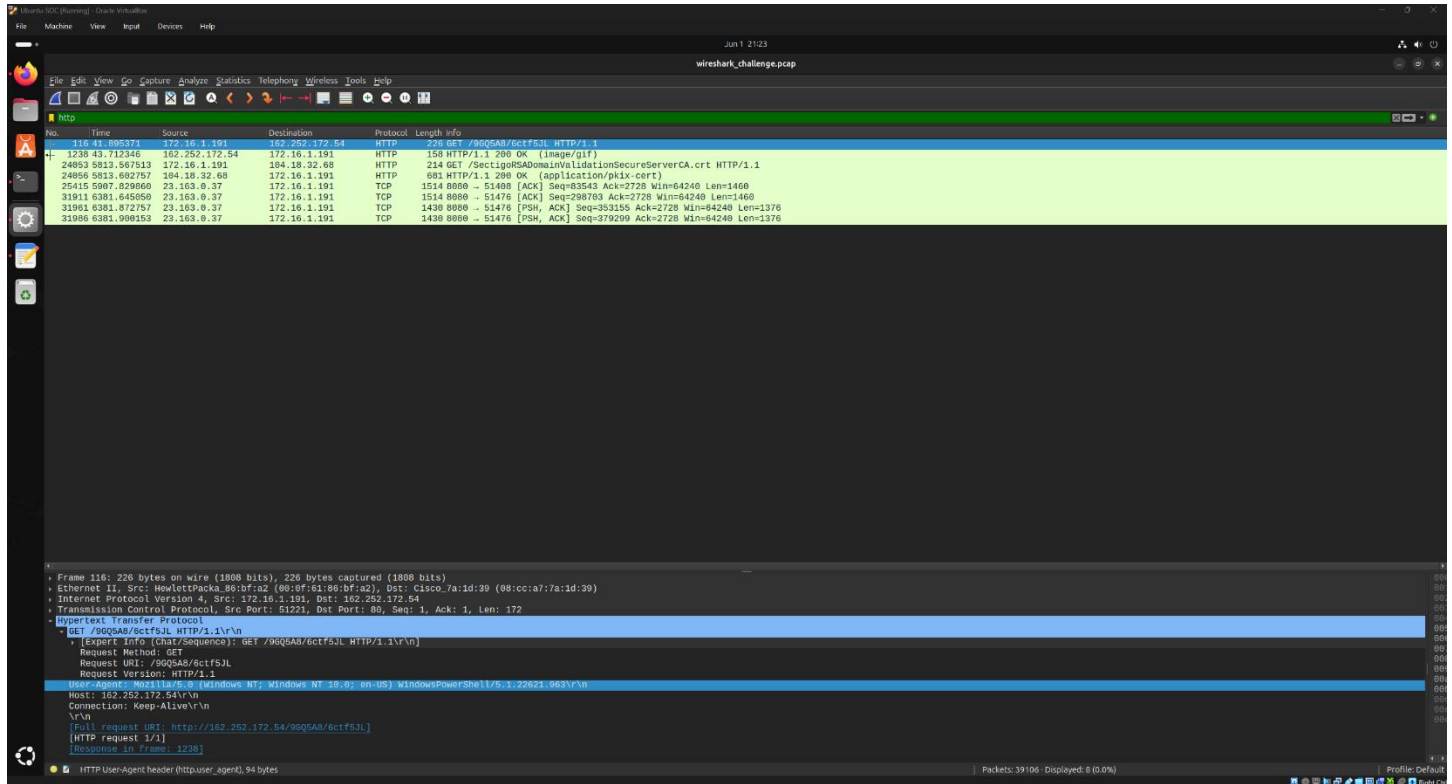


3.4 Execution & File Identification

Q8. What command-line utility or program was used by the victim to download the file?

Method / Filter: Examined the HTTP request's User-Agent header

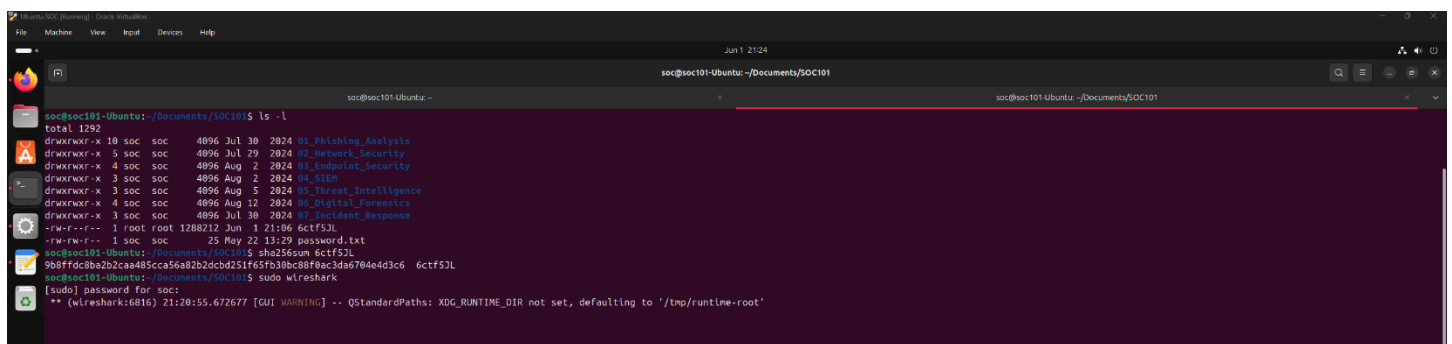
Finding: WindowsPowerShell — indicating the file was downloaded via a PowerShell command rather than a standard browser, a common technique for living-off-the-land malware delivery.



Q9. What is the SHA256 hash of the downloaded file?

Method / Filter: Exported the object from Wireshark (File → Export Objects → HTTP), then hashed it locally using sha256sum

Finding: 9b8ffdc8ba2b2caa485cca56a82b2dcbd251f65fb30bc88f0ac3da6704e4d3c6



Q10. Submit the uncovered hash to VirusTotal. Based on the popular threat label and tags, what type of malware did the endpoint get infected with?

Method / Filter: VirusTotal hash lookup using the SHA256 value above

Finding: Trojan/Win.Qakbot — VirusTotal's community threat label identified the file as a variant of the Qakbot banking trojan.

59/71 security vendors flagged this file as malicious

9b0ffdc8b2ca485ca5a82b2d0bd251f63fb30bc88f0ac3da9704e4d3ce/detection

Size: 1.2 MB | Last Analysis Date: 11 hours ago

Popular threat label: Trojan:Win32/Zenpak

Threat categories: Trojan

Family labels: Zenpak, Win32, Trojan

Security vendors' analysis:

Vendor	Detection	Family
AhnLab-V3	Trojan:Win32/Zenpak-1579273	Trojan:Win32/Zenpak-4295888
AllCloud	Trojan:Win32/Zenpak-1579273	Trojan:Agent/Pakbot
Antiy-AVL	Trojan:Win32/Zenpak	Trojan:Win32/Zenpak
Arctic Wolf	Unlabeled	Win32/Malware-gen (Troj)
AVG	Win32/Malware-gen (Troj)	Win32/Malware-gen
BitDefender	GenVariant/Malware.147122	Win32/Malware.DCFE1084
ClamAV	Win.Loader.Bkbl-13007396-0	Win32/Malware.confidence_100% (W)
CTX	Dll Trojan generic	Win32/Malware (score: 100)
DeepInstinct	Malicious	Trojan:Win32/Zenpak
Elastic	Malicious (High Confidence)	GenVariant/Malware.147122 (B)
escan	GenVariant/Malware.147122	Win32/Malware (score: 100)
Fortinet	Win32/Malware-gen (Troj)	Win32/Malware (score: 100)
Google	Detected	Win32/Malware (score: 100)
Jiangmin	Trojan:Win32/Zenpak	Win32/Malware (score: 100)
K7Win	Trojan (00560341)	Win32/Malware (score: 100)
K7Win	Trojan (00560341)	Win32/Malware (score: 100)

3.5 Post-Infection Network Behavior

Q11. What protocol makes up the majority of UDP packets?

Method / Filter: Statistics → Protocol Hierarchy

Finding: DNS (Domain Name System) accounts for the majority of UDP traffic in the capture.

Wireshark - Protocol Hierarchy Statistics - wireshark_challenge.pcap

Protocol	Percent Packets	Packets	Percent Bytes	Bytes	Bits/s	End Packets	End Bytes	End Bits/s	PDU/s
Frame	100.0	39106	100.0	11146724	9.603	0	0	0	39106
Ethernet II	100.0	39106	100.0	11146724	9.603	0	0	0	39106
Internet Protocol Version 4	100.0	39106	100.0	11146724	9.603	0	0	0	39106
User Datagram Protocol	78.2	29784	2.1	238772	2.05	0	0	0	29784
Transmission Control Protocol	23.8	9322	48.4	5355280	4.546	0	0	0	9322
Transport Layer Security	0.6	329	2.7	297947	2.55	329	243968	210	342
Hypertext Transfer Protocol	0.6	329	11.6	1288212	1.109	1	1288212	1,109	1
Media Type	0.0	1	11.6	1288212	1.109	1	1288212	1,109	1

Packet 10 Details:

- Ethernet II, Src: Intel(R) Ethernet Controller (08:00:27:00:00:00), Dst: 172.16.1.191
- Internet Protocol Version 4, Src: 172.16.1.191, Dst: 172.16.1.191
- User Datagram Protocol, Src Port: 53, Dst Port: 53
- DNS Standard query response, Transaction ID: 0x00000000, Flags: 0x0000, QID: 0x0000, Qtype: 0x0000, Qclass: 0x0000, Rcode: 0x0000, Answer: 0x0000, Authority: 0x0000, Additional: 0x0000

Q12. Look at the domain names that were queried within the capture. In defanged format, what is the base domain name that is continually queried?

Method / Filter: Reviewed repeated DNS query patterns across the capture

Finding: aaa[.]h[.]dns[.]steasteel[.]net — queried repeatedly with varying subdomains, a strong indicator of DNS tunneling for C2 communication.

Q13. Read up on MITRE ATT&CK ID T1071.004. What is the attack technique we are likely seeing in the PCAP file often known as?

Method / Filter: MITRE ATT&CK framework reference, technique T1071.004 — Application Layer Protocol: DNS

Finding: DNS Tunneling — the malware is encoding data within DNS queries/responses to establish covert command-and-control communication, evading traditional network defenses that focus on HTTP/HTTPS traffic.

The screenshot shows the MITRE ATT&CK framework website. The left sidebar lists various techniques under categories like Discovery, Lateral Movement, Collection, Command and Control, etc. The main content area is titled 'Application Layer Protocol: DNS'. It includes a description of the technique, a list of other sub-techniques, and a table of procedure examples.

Other sub-techniques of Application Layer Protocol (5)

- Adversaries may communicate using the Domain Name System (DNS) application layer protocol to avoid detection/network filtering by blending in with existing traffic. Commands to the remote system, and often the results of those commands, will be embedded within the protocol traffic between the client and server.
- The DNS protocol serves an administrative function in computer networking and thus may be very common in environments. DNS traffic may also be allowed even before network authentication is completed. DNS packets contain many fields and headers in which data can be concealed. Often known as **DNS tunneling**, adversaries may abuse DNS to communicate with systems under their control within a victim network while also mimicking normal, expected traffic.^{[1][2]}
- DNS beaconing may be used to send commands to remote systems via DNS queries. A DNS beacon is created by tunneling DNS traffic (i.e. Protocol Tunneling). The commands may be embedded into different DNS records, for example, TXT or A records.^[3] DNS beacons may be difficult to detect because the beacons infrequently communicate with infected devices.^[4] Infrequent communication conceals the malicious DNS traffic with normal DNS traffic.

Procedure Examples

ID	Name	Description
S0504	Anchor	Variants of Anchor can use DNS tunneling to communicate with C2. ^[5]
G0026	APT18	APT18 uses DNS for C2 communications. ^[1]
G0087	APT39	APT39 has used remote access tools that leverage DNS in communications with C2. ^[6]
G0096	APT41	APT41 used DNS for C2 communications. ^{[6][7]}
S0360	BONDUPDATER	BONDUPDATER can use DNS and TXT records within its DNS tunneling protocol for command and control. ^[1]
S9015	BRICKSTORM	BRICKSTORM has used DNS over HTTPS to resolve C2 infrastructure and obscure DNS traffic from inspection. ^{[8][9][10][11]}
S1063	Brute Ratel C4	Brute Ratel C4 can use DNS over HTTPS for C2. ^{[12][13]}
G0114	Chimera	Chimera has used Cobalt Strike to encapsulate C2 in DNS traffic. ^[14]
G0080	Cobalt Group	Cobalt Group has used DNS tunneling for C2. ^{[15][16][17]}
S0154	Cobalt Strike	Cobalt Strike can use a custom command and control protocol that can be encapsulated in DNS. All protocols use their standard assigned ports. ^{[18][19][20]}
S0338	Cobian RAT	Cobian RAT uses DNS for C2. ^[21]

Metadata:

- ID: T1071.004
- Sub-technique of: T1071
- Tactic: Command and Control
- Platforms: ESX, Linux, Network Devices, Windows, macOS
- Contributors: Chris Heald, Jan Petrov, Citi
- Version: 1.4
- Created: 15 March 2020
- Last Modified: 12 May 2026

MITRE ATT&CK Mapping

Technique ID	Tactic	Observed Behavior
T1071.004	Command and Control	Repeated DNS queries to steasteel.net subdomains, consistent with DNS tunneling for covert C2 channel
T1105	Command and Control	Ingress Tool Transfer — malicious PE executable downloaded over HTTP, disguised with an image/gif Content-Type header
T1059.001	Execution	PowerShell identified as the User-Agent / download utility, indicating a living-off-the-land execution technique

5. Conclusion & Recommendations

5.1 Summary of Compromise

Host 172.16.1.16 was compromised after a user (or automated process) executed a PowerShell command to download a file from a remote web server. Although the server's HTTP response header falsely declared the file as an image/gif, byte-level inspection revealed the MZ magic bytes of a Windows executable. The hash of this file (9b8ffdc8ba2b2caa485cca56a82b2dcdbd251f65fb30bc88f0ac3da6704e4d3c6) was confirmed via VirusTotal to be a variant of the Qakbot banking trojan. Following execution, the host began issuing a high volume of DNS queries to a single base domain (steasteel.net) with varying subdomains — a pattern strongly associated with DNS tunneling, used here for covert command-and-control.

5.2 Recommendations

1. Isolate the affected endpoint (172.16.1.16) from the network immediately to prevent lateral movement and further data exfiltration.
2. Block the malicious domain (steasteel.net and subdomains) and the delivery server IP (184.168.98.64) at the firewall/DNS resolver level.
3. Deploy detection rules for DNS query volume anomalies (e.g., high query rate to a single base domain) to catch similar tunneling attempts in the future.
4. Restrict PowerShell execution policies and enable script block logging to detect living-off-the-land download techniques.
5. Conduct a full forensic sweep of the affected host for persistence mechanisms (scheduled tasks, registry run keys, services) commonly used by Qakbot.
6. Submit the malicious file hash to internal threat intelligence feeds and EDR allow/deny lists across the environment.

5.3 Skills Demonstrated

- Network traffic analysis using Wireshark display filters and statistics
- HTTP stream reconstruction and file carving
- File signature / magic byte verification (defeating false Content-Type headers)
- Cryptographic hashing for file identification (SHA256)
- Threat intelligence correlation via VirusTotal
- DNS tunneling detection and MITRE ATT&CK technique mapping